The Empirical Model in *photic*: Lyzenga's Log-Linear Fit, Diffuse Attenuation and Depth Errors

Technical summary

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Abstract

`MODEL lyzenga_log_linear` fits Lyzenga's log-linear depth model to soundings, such as ICESat-2 bathymetric photons. The model can be fitted to one image or to several scenes at once. This note sets out how *photic* makes three estimates.

1. The depth, and its 1σ prediction error (`ERROR`), from a robust weighted least-squares fit.
2. The diffuse attenuation coefficient K of every scene and band, as a smooth field over the scene (`ATTENUATION`, stage 1). It is estimated at the soundings, using the physical model that Lyzenga's regression linearises.
3. A physical depth at every cell, from how the attenuation differs between bands and scenes, with its own error estimate (stage 2).

All of it is implemented in C, in `model/lyzenga.c`. The results quoted are from the Arabian Gulf validation site between Dubai and Abu Dhabi: six Landsat 8 scenes, against ICESat-2 ATL24 split by pass into calibration and test halves.

Lyzenga's fit across all scenes jointly scores 90.0% hits on the test half. Its error estimate holds 80% of the errors within $\pm1\sigma$, but its tails are heavier than Gaussian. K at 443–561 nm is plausible and spatially coherent. K at 655 nm falls below pure water's absorption, so it is not physical. The depth from the attenuation is a diagnostic, not a product, and its error estimate is too small by a factor of about two to three.

Lyzenga et al.'s calibration condition (their eq. 12) explains where the fit is unbiased. The stage-1 attenuation shows that the free fit meets it only at 5–10 m, and that the saturation near 16 m is the condition failing as $\ln L$ flattens. Imposing it on the fit changes almost nothing. Dividing each predictor by the local attenuation (`NORMALISE`) gains about 3 points of hits on the same cells, nearly all at 10–20 m, but loses coverage, so the chosen model is unchanged.

Lyzenga's 1978 paper adds two things. Dividing the error by the local gain (its eq. 17) makes σ grow where the fit saturates: beyond 15 m, 87% of the test errors now lie within $\pm1\sigma$, against 23%. Its depth-invariant bottom indices, computed with the local attenuation, remove most of the depth from the signal but not all, and do not yet give a consistent bottom map at the Gulf.

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1 Notation and inputs

Grids. The command takes n band grids L_m, $m = 1, \ldots, n$, all on one grid. They can be remote-sensing reflectance from ACOLITE, or radiance. With `ATTENUATION ... SCENES` S, the grids are scene-major: S scenes of $n_b = n/S$ bands each, in the same band order. Grid m belongs to scene $s(m) = \lfloor m/n_b \rfloor$.

Window means. Each band is averaged over a $(2r-1) \times (2r-1)$ window of valid cells (`RADIUS` r), giving $\bar{L}_m(\mathbf{x})$. A cell is modelled only where every band is valid. `LAND` and `SHALLOW` masks, if given, further restrict it.

Deep-water signal and noise. For each band:

- $L_{\infty,m}$ is either given (`LSM`) or the p-th percentile of the window means (`PERCENTILE`, default $p = 1$).
- σ_m is the robust spread of the darkest cells: 1.4826 times the median absolute deviation of the window means at or below that percentile. It stands for the noise of a deep-water cell.

The signal above deep water is

$$v_m(\mathbf{x}) = \bar{L}_m(\mathbf{x}) - L_{\infty,m}. \tag{1.1}$$

A band sees the bottom while $v_m \geq k\,\sigma_m$ (`DEEP` k, default 2). A cell where any band fails this test is optically deep for the Lyzenga fit, and gets no depth.

Soundings and tides. Soundings z are depths below the datum, positive down. Each sounding goes to its nearest cell; only soundings within `DEPTHS` $[z_{\min}, z_{\max}]$ are used. `TIDE` h is the water level above the datum when the image was taken, so the light saw a water column $Z = z + h$. With several scenes, scene s has its own level h_s (`TIDES`), and $Z_s = z + h_s$. Outputs are depths below the datum, negative down.

The soundings file must hold depths below the datum. If the tide is added both to the file and through `TIDE`, it is counted twice; this happened in the Gulf composite runs and was fixed on 20 October 2026.

2 Lyzenga's regression

2.1 The model and its fit

With $X_m = \ln v_m$, the model (Lyzenga 1978, 1985; Lyzenga et al. 2006) is

$$\hat{Z} = a_0 + \sum_{m=1}^{n} a_m X_m = \mathbf{a}^{\mathsf{T}}\mathbf{x}, \qquad \mathbf{x} = (1, X_1, \ldots, X_n)^{\mathsf{T}}. \tag{2.1}$$

Setting `LSM 0` gives the plain $\ln L$, with no deep-water signal subtracted; Lyzenga et al. (2006) always subtract one. Fitted to every scene's bands at once, equation (2.1) is the *joint* model. It includes every weighted sum of per-scene fits as a special case. The scenes' tides are constants per scene, so they go into the coefficients, and the target is depth below the datum (`TIDE 0`).

The rows are the N soundings whose cells see the bottom in every band. The coefficients minimise

$$\sum_{i=1}^{N} w_i \big(Z_i - \mathbf{a}^{\mathsf{T}}\mathbf{x}_i\big)^2, \tag{2.2}$$

solved by Householder QR of $W^{1/2}A$, which gives the triangular factor R with $R^{\mathsf{T}}R = A^{\mathsf{T}}WA$. The weights are $w_i = w_i^0\,\psi_i$:

- w_i^0 is 1, or 1/(count in the sounding's depth bin) with `BALANCE`.
- ψ_i are Huber weights, from iteratively reweighted least squares: $\psi_i = \min\big(1,\ k_H s/|e_i|\big)$, where e_i is the residual, $s = 1.4826\,\mathrm{median}|e_i|$, and $k_H = 1.345$ (`HUBER`). The iterations start from the least-squares solution.

2.2 Rescaling against regression dilution (`POWER`)

Noise in the predictors shrinks the fitted depths towards the mean, so deep water reads too shallow. `POWER` uses classical calibration to correct this:

1. At the soundings, the fit's depths are modelled as $\hat{Z} = c\,Z^d$, fitted robustly in logs: $\ln\hat{Z} = \ln c + d\ln Z$, with the same Huber IRLS.
2. The law is inverted: $Z' = (\hat{Z}/c)^{1/d}$.

By default, the inverse is applied only beyond the crossover $\hat{Z}_\times = c^{1/(1-d)}$. That is where the fit is compressed into too little depth, so shallow depths keep the fit's smaller scatter. `POWER ALL` applies the inverse at every depth. At the Gulf, the joint fit gives $\hat{Z} = 1.081\,Z^{0.959}$, with the crossover at 6.9 m.

2.3 The depth error estimate (`ERROR`)

The `ERROR` grid is the 1σ prediction error of a new depth at cell $\mathbf{x}$. Write W for the final weights, rescaled to average 1 ($\tilde{w}_i = N w_i / \sum_j w_j$). The residual variance of a sounding of average weight is

$$s^2 = \frac{1}{N-p} \sum_{i=1}^{N} \tilde{w}_i \, e_i^2, \qquad p = n + 1. \tag{2.3}$$

The coefficients' covariance follows from the QR factor:

$$C = s^2 \big(A^{\mathsf{T}} \tilde{W} A\big)^{-1} = s^2 \, \frac{\sum_j w_j}{N} \, (R^{\mathsf{T}} R)^{-1}. \tag{2.4}$$

The prediction variance at a cell with predictors $\mathbf{x}$ is then

$$\sigma^2_{\hat{Z}}(\mathbf{x}) = s^2 + \mathbf{x}^{\mathsf{T}} C \, \mathbf{x}. \tag{2.5}$$

It has two parts: the scatter of one sounding about the model, and the uncertainty of the coefficients, which grows as $\mathbf{x}$ moves away from the soundings' centre.

Through `POWER`, a small error in $\hat{Z}$ scales by $\mathrm{d}Z'/\mathrm{d}\hat{Z} = Z'/(d\,\hat{Z})$. So where the rescaling applies,

$$\sigma_{Z'} = \sigma_{\hat{Z}} \, \frac{Z'}{d\,\hat{Z}}. \tag{2.6}$$

What it represents. Because Huber weighting is used, s^2 in (2.3) is a robust variance. Outlying soundings are down-weighted, so s (0.70 m at the Gulf) describes the core of the residuals, not their tails (in-sample rms 1.86 m). The estimate assumes

- residuals independent between soundings, with one variance at every depth and place;
- the model's form is right.

It therefore cannot see:

- saturation where the bands stop seeing the bottom;
- spatially correlated error, for example from a change of bottom type;
- errors in the soundings themselves.

Noise and gain (`NOISE`, `GAIN`). Lyzenga (1978) derives the depth error from the sensor noise and the signal's sensitivity to depth C (his eqs. 16 and 17):

$$(\Delta Y_N)^2 = \sum_i A_{Ni}^2 \frac{\mathrm{NE}\Delta L_i^2}{(L_i - L_{si})^2}, \qquad \Delta Z = \frac{\Delta Y_N}{C}. \tag{2.7}$$

Two options of `ERROR` follow from them.

- `NOISE` adds each cell's own noise, $n^2(\mathbf{x}) = \sum_m a_m^2 \sigma_m^2 / (\bar{L}_m - L_{0,m})^2$, where $L_{0,m}$ is the signal subtracted in the fit (0 for $\ln L$). The soundings' mean of it, $\bar{n}^2$, is already part of s^2, so it is taken out first: $\sigma^2 = \max(s^2 - \bar{n}^2, 0) + n^2(\mathbf{x}) + \mathbf{x}^{\mathsf{T}} C \mathbf{x}$.

- `GAIN` (with `ATTENUATION`) divides σ by the local gain of the fitted depth on the true depth. This is C, and it is the gain of Section 7:

$$G(\mathbf{x}) = -\sum_m a_m\, g_m(\mathbf{x})\, f_m, \qquad f_m = \min\Big(1, \frac{\bar{L}_m - L_{\infty,m}}{\bar{L}_m - L_{0,m}}\Big), \tag{2.8}$$

  where $L_{\infty,m}$ is the deep-water signal (the percentile), so $f_m = 1$ for $\ln(L - L_\infty)$ and falls towards deep water for $\ln L$. G needs no depth: it comes from the observed signal and the K maps. Where the fit saturates, G falls and σ grows. G is capped at 1, so the shallow water keeps the fit's own scatter, and floored at 0.05. With `POWER`, the larger of its rescaling and $1/G$ is used, as both undo the same compression.

Calibration at the Gulf. This is the joint model, four bands per scene, with `POWER`, scored against ICESat-2 cells that the fit did not use (Table 1).

- **Plain `ERROR`:** 80% of the errors lie within $\pm1\sigma$ and 93.6% within $\pm2\sigma$ (68 and 95% for a Gaussian), and the rms of e/σ is 1.6. So the core is conservative and the tails are heavy. σ is 0.70–0.76 m at every depth, because $\mathbf{x}^\mathsf{T} C \mathbf{x}$ is small next to s^2 for 6,017 soundings. It misses the saturation entirely: beyond 15 m, only 23% of the errors are within $\pm1\sigma$.
- **With `NOISE GAIN`:** 86.9% lie within $\pm1\sigma$ and 96.4% within $\pm2\sigma$, and the rms of e/σ is 1.14. At 15–20 m, 87% are within $\pm1\sigma$, with a median σ of 3.1 m. Nothing changes shallower than 5 m. The noise itself is small, 0.13–0.18 m of the fit's 0.65–0.75 m; the gain does the work.
- **Out of fold** (the four cross-validation folds, calibration half), the same holds: the share within $\pm1\sigma$ beyond 15 m goes from 9 to 81%, and the rms of e/σ from 2.9 to 1.6. The two forms of the gain (capped at 1 or not) were compared on the test half first, so this is the confirmation.
- **Still too small at 20–30 m** (22%): there the depth no longer responds at all, and the error is most of the depth.

So with `NOISE GAIN`, σ is a fair guide to the error at every depth the model reaches, and it flags where it stops reaching. It is still not a bound.

Table 1: Calibration of Lyzenga's `ERROR` on the Gulf's test half (joint model, four bands, `POWER`; 5,543 cells with a depth), plain and with `NOISE GAIN`.

		`ERROR`		`ERROR NOISE GAIN`		
Depth (m)	Cells	Within $\pm1\sigma$	Median σ	Within $\pm1\sigma$	Median σ	RMSE (m)
0–2	161	83%	0.70	83%	0.69	2.38
2–5	1,352	87%	0.70	87%	0.69	1.32
5–10	2,027	79%	0.74	82%	0.75	0.78
10–15	1,737	84%	0.75	94%	0.96	0.65
15–20	243	23%	0.76	87%	3.08	2.15
20–30	23	0%	0.76	22%	3.39	8.77
All	5,543	80%	0.74	87%	0.75	1.21

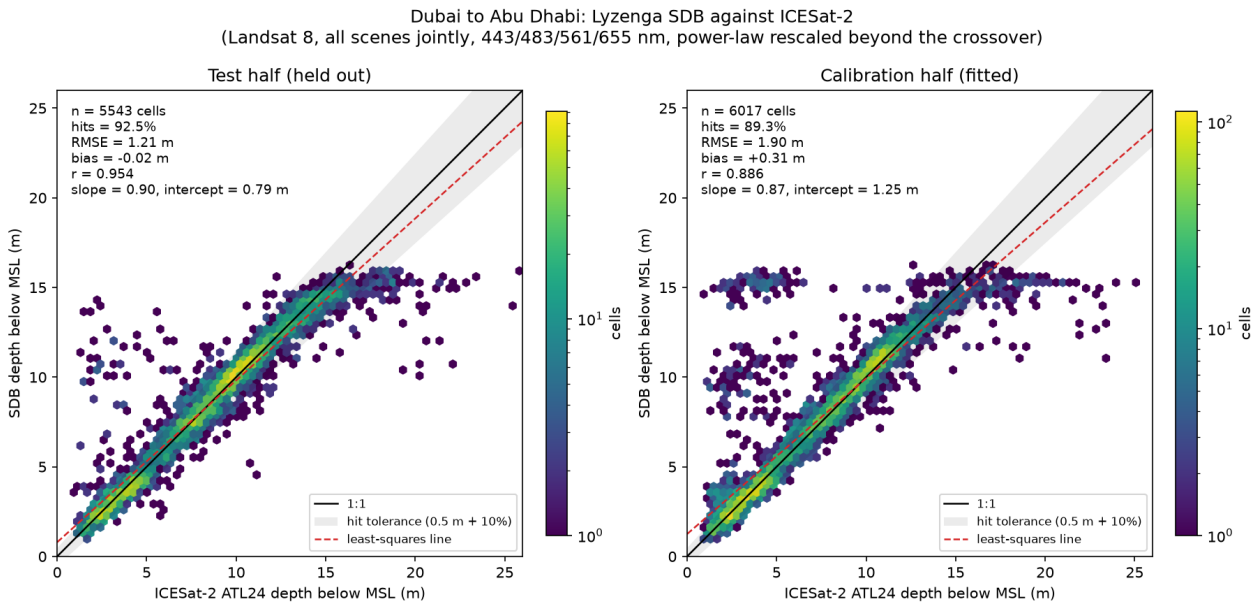


Figure 1: The joint Lyzenga model against ICESat-2 at the Gulf. The test half is held out by pass; the calibration half was fitted. Hits here count cells with a depth. Counting every reference cell, the test-half score is 90.0%.

3 The attenuation behind the regression

Lyzenga's model is the linearisation of a two-flow model of the water-leaving signal over an optically shallow bottom:

$$L_m - L_{\infty,m} \;\approx\; A(\mathbf{x})\, r_m \,\exp\!\big(-g_m(\mathbf{x})\, Z_s\big), \qquad g_m = K_m\, M_s, \tag{3.1}$$

where

- $A(\mathbf{x})$ is the bottom's brightness, taken as spectrally flat in logs (Lyzenga's assumption);
- r_m is the band's bottom colour, together with the band's deep-water and surface constants;
- K_m is the effective diffuse attenuation of the band;
- M_s is the path length in water per metre of depth, down and back up.

M_s comes from the sun and view zeniths in air (`SUNZENITH`, `VIEWZENITH`), refracted into water with $n_w = 1.34$:

$$M_s = \sec\theta_{w,\mathrm{sun}} + \sec\theta_{w,\mathrm{view}}, \qquad \sin\theta_w = \sin\theta / n_w. \tag{3.2}$$

With both zeniths at 0, $M_s = 2$. For the Gulf scenes (sun zenith 35–51°, near-nadir view), M_s is about 2.1–2.25.

If the down- and up-welling attenuations differ, $g_m = K_d \sec\theta_{w,\mathrm{sun}} + K_u \sec\theta_{w,\mathrm{view}}$. The reported $K_m = g_m/M_s$ is then their path-weighted mean.

Lyzenga (1978, eq. 6) shows that scattering in the water column has the same depth dependence as the bottom's signal. So $A\,r_m$ is an *apparent* bottom reflectance,

$$r'_B = r_B - \frac{\pi\beta(\mu_s)}{K(\mu+\mu')}, \tag{3.3}$$

which depends on the water as well as the bottom (β is the volume scattering function, and μ, μ' the cosines of the paths). A bottom darker than the water column's own scattering has $r'_B < 0$: its signal lies below deep water's, $L < L_\infty$, and its cell is dropped. The same paper notes the model's remaining nonlinearity: internal reflection at the surface bends $\ln(L - L_\infty)$ against depth in very shallow water over bright bottoms.

Taking logs of (3.1) gives, for grid m of scene s,

$$X_m = \ln v_m = B(\mathbf{x}) + c_m - g_m(\mathbf{x})\, Z_s, \qquad B = \ln A,\ c_m = \ln r_m. \tag{3.4}$$

With g_m and c_m constant and B constant, Lyzenga's regression (2.1) is this equation solved for Z. In practice g_m varies over a scene, and between scenes, with the water. `ATTENUATION` estimates it explicitly, in two stages.

In stage 1, $L_{\infty,m}$ is the `LSM` value where that is positive, and otherwise the percentile. So `LSM 0` (the plain $\ln L$) still subtracts a deep-water signal for the attenuation.

4 Stage 1: the attenuation coefficients at the soundings

4.1 A smooth field for each grid

At the soundings the depth is known, so for each grid m separately, equation (3.4) is fitted with B left in the residuals:

$$X_{m,i} = c_m - g_m(\mathbf{x}_i)\, Z_{s,i} + \varepsilon_{m,i}. \tag{4.1}$$

$g_m(\mathbf{x})$ is a bilinear interpolant of values $\theta_{m,q}$ on a regular grid of nodes, set `NODES` degrees apart (default 0.05°; 12×9 nodes over the Gulf box):

$$g_m(\mathbf{x}) = \sum_{q=1}^{4} f_q(\mathbf{x})\, \theta_{m,q(\mathbf{x})}, \tag{4.2}$$

where $q(\mathbf{x})$ are the four nodes around the cell and f_q their bilinear weights. The unknowns, c_m and the Q node values $\theta_{m,\cdot}$, minimise

$$\sum_i w_{m,i}\Big(X_{m,i} - c_m + Z_{s,i} \sum_q f_q(\mathbf{x}_i)\theta_{m,q}\Big)^2 \;+\; \lambda \sum_{(q,q')} \big(\theta_{m,q} - \theta_{m,q'}\big)^2. \tag{4.3}$$

The second sum runs over neighbouring pairs of nodes, east–west and north–south. It is a first-difference smoothing penalty, which makes the field smooth and fills nodes without soundings. Its weight is scaled to an average node's data:

$$\lambda = w_s\, \frac{\sum_i w^0_{m,i} Z^2_{s,i}}{Q}, \tag{4.4}$$

where w_s is `SMOOTHING` (default 1; 0 requires soundings near every node). The normal equations, of size $Q + 1$, are symmetric positive definite and are solved by Cholesky factorisation. The residuals are robustified by the same Huber IRLS as in Section 2, with scale $1.4826\,\mathrm{median}|\varepsilon|$.

4.2 Weights from the noise

The noise σ_m on v_m gives $\ln v_m$ a variance of about σ_m^2/v_m^2 (delta method). This grows without limit as the signal sinks towards the deep-water level. A floor $\epsilon = 0.05$ stands for the model's own error in $\ln v$. The weights are

$$w^0_{m,i} = \Big(\frac{\sigma_m^2}{v_{m,i}^2} + \epsilon^2\Big)^{-1}, \tag{4.5}$$

so a band counts fully while it sees the bottom clearly, and fades out as it loses it. The Huber factors multiply w^0.

4.3 Selecting soundings by the fitted signal

A sounding enters grid m's fit only while its signal is above the noise, $v \geq k\sigma_m$. Selecting, and weighting, by the *observed* signal truncates the data on the dependent variable. At depth, only the cells that happen to be bright pass the test, which flattens the slope and makes K too small. The effect is largest in the red, which loses the bottom within a few metres.

So after the first pass, each candidate sounding is reselected and reweighted by the *fitted* signal instead:

$$\hat{v}_{m,i} = \exp\big(c_m - g_m(\mathbf{x}_i)\, Z_{s,i}\big), \qquad \text{active if } \hat{v}_{m,i} \geq k\sigma_m, \tag{4.6}$$

with $\hat{v}$ in place of v in (4.5). The fit is repeated until the selection stops changing, for at most 10 passes. This raised $K(655)$ at the Gulf by 20–40%. A grid needs at least 10 active soundings.

4.4 Outputs

- $K_m(\mathbf{x}) = g_m(\mathbf{x})/M_{s(m)}$ at every valid cell, as grid `/prefix/_/grid/`.
- `attenuation.csv`: K of every grid at every node, with the node's support (the sum of the bilinear weights of the soundings near it).
- The printout gives, for each grid: c_m, the median K over nodes with support of at least 1, its range, the robust scatter of (4.1), and the number of passes.

Caveat. B is left in the residuals. K is therefore unbiased only where the bottom's brightness does not change systematically with depth across the soundings. If seagrass is deeper than sand, the slope also contains the brightness change.

5 Stage 2: a depth from the attenuation

5.1 A straight line at every cell

With c_m and $g_m(\mathbf{x})$ fixed from stage 1, equation (3.4) at a cell, over all its grids, is a straight line in g_m:

$$y_m \equiv X_m - c_m + g_m(\mathbf{x})\, h_{s(m)} = B(\mathbf{x}) - g_m(\mathbf{x})\, H(\mathbf{x}), \tag{5.1}$$

where H is the depth below the datum. The intercept B is the bottom's brightness and the slope is $-H$. The depth comes from how g differs between bands and scenes, which is what makes it independent of B.

The weighted least-squares line through the grids that see the bottom gives

$$\hat{H} = -\frac{S_{gy}}{S_{gg}}, \qquad \hat{B} = \bar{y} + \hat{H}\bar{g}, \tag{5.2}$$

with

$$\bar{g} = \frac{\sum w_m g_m}{\sum w_m}, \quad \bar{y} = \frac{\sum w_m y_m}{\sum w_m}, \quad S_{gg} = \sum_m w_m (g_m - \bar{g})^2, \quad S_{gy} = \sum_m w_m (g_m - \bar{g})(y_m - \bar{y}). \tag{5.3}$$

The weights are (4.5) with the cell's signal. As in stage 1, the first pass selects and weights by the observed signal; later passes use the fitted signal,

$$\hat{v}_m = \exp\big(\hat{B} + c_m - g_m(\hat{H} + h_{s(m)})\big) = v_m \exp\big(\hat{B} - g_m\hat{H} - y_m\big), \tag{5.4}$$

until the selection settles (at most 10 passes). A cell needs at least 3 grids above $k\sigma_m$.

The outputs are three grids:

- `/prefix/_depth`, the depth $-\hat{H}$ (negative down);
- `/prefix/_depth_sigma`, its error σ_H;
- `/prefix/_bottom`, the bottom brightness $\hat{B}$.

5.2 The depth error estimate

σ_H is the classical standard error of the slope of a weighted straight line, with the residual scale estimated from the cell's own scatter about the line:

$$\sigma_H = \sqrt{\frac{\mathrm{RSS}}{(n_u - 2)\, S_{gg}}}, \qquad \mathrm{RSS} = \sum_m w_m \big(y_m - \hat{B} + g_m\hat{H}\big)^2, \tag{5.5}$$

where n_u is the number of grids used.

The leverage is S_{gg}, the weighted spread of the two-way attenuation among the cell's grids. It is small. Between 443 and 561 nm, K differs by only 0.01–0.03 m^{-1}, so g spans about 0.02–0.07 m^{-1}, and the scenes' different sun angles add little. A bottom colour that departs from spectral flatness by δ in ln therefore moves the depth by about $\delta/\Delta g$. For $\delta = 0.1$ that is 1.5–5 m, comparable to the depths being estimated. This is Lyzenga's (1978, eq. 18) result for the two-band ratio method, whose noise error is $\left[\sum_i (\mathrm{NE}\Delta L_i/(L_i - L_{si}))^2\right]^{1/2}/((\kappa_1 - \kappa_2)f)$ and grows without limit as the bands' attenuations approach each other.

Equation (5.5) treats the n_u residuals as independent, with a common scale. It omits three things.

1. **Stage 1's uncertainty.** c_m and $g_m(\mathbf{x})$ are taken as exact. Their errors are shared by every cell near the same nodes, so they are systematic, not random, and they do not average down.
2. **The bottom's colour.** A non-flat bottom spectrum gives each band its own offset in y_m. The same offset recurs in every scene, so the residuals are correlated across scenes, and with only a few distinct bands it can be fitted into the slope rather than showing up as scatter.
3. **Few effective degrees of freedom.** The 18–24 grids are only 3–4 distinct bands seen 6 times under similar water. $n_u - 2$ overstates the information.

All three make σ_H too small. Out of fold at the Gulf, only 25% (four bands) and 37% (443, 483 and 561 nm) of the errors lie within $\pm\sigma_H$, against 68% for a calibrated estimate.

Two corrections are possible:

- propagate stage 1's covariance (the Cholesky factor is at hand) into (5.5);
- inflate σ_H by a factor calibrated out of fold.

Neither is built.

6 Results at the Gulf

6.1 The attenuation coefficients

Over the box, K is 0.05–0.09 m^{-1} at 443 nm, 0.07–0.10 at 483 and 0.065–0.105 at 561 (medians over the scenes), and it varies 1.5–2× across the box (Figure 3). The three bands are fitted independently, yet they map the same pattern: low in the south-west, rising towards Jebel Ali and Dubai. The scenes differ: 2019-03-13 is the clearest, 2018-10-04 the least clear. One set of Lyzenga coefficients cannot represent this variation. These maps are the per-site attenuation targets that the learned model's physics decoder needs (`validation/FUTURE_EXPERIMENTS.md`, H4(c)).

Pure water's absorption a_w (Pope and Fry 1997) is a lower bound on K (Figure 2).

- **443 and 483 nm:** K is 4–15 times a_w, as expected where dissolved and particulate matter dominate.
- **561 nm:** K is barely above a_w (0.065–0.105 against 0.062 m^{-1}). 6% of the node values fall below it, most of them in the clearest scene.
- **655 nm:** every value is below a_w (0.08–0.19 against about 0.37 m^{-1}), so none is physical.

The red loses the bottom within a few metres. Its deep-water signal varies (turbidity, adjacency, the atmospheric residual) by far more than the darkest cells' spread σ_m, which sets the threshold. So nearly every sounding stays active, and the slope is flattened. The flattening already reaches the green on clear scenes.

Two remedies are possible, and neither is built:

- fit the deep-water level per grid, or as a field;
- bound K below by a_w.

A pure-water bound would bind at 561 nm as well as at 655.

6.2 The depth from the attenuation

Table 2 scores the depth from the attenuation against Lyzenga's depth, out of fold. The folds are the four ICESat-2 pass groups that chose Lyzenga's settings, within the calibration half only. The harness reproduces Lyzenga's cross-validation score (83.9%).

The depth from the attenuation is not competitive with Lyzenga's. With so little spread in g, a change of bottom type (seagrass against sand) shifts the band ratios enough to move the depth by metres. Lyzenga's regression absorbs the site's bottom–depth relationships into its coefficients; equation (5.1) cannot. Dropping the red removes the unphysical K and helps the shallowest water

Table 2: Out of fold on the Gulf's calibration half (5,833 cells). A hit is an error within 0.5 m + 10% of the depth.

	Hits	RMSE (m)	Bias (m)	0–2	2–5	5–10	10–20
Lyzenga, joint, four bands	83.9%	2.09	+0.26	58%	79%	90%	90%
Attenuation depth, four bands	57.3%	3.14	−0.58	5%	56%	79%	47%
Attenuation depth, 443/483/561	61.4%	2.84	+0.12	34%	54%	69%	69%

most (0–2 m: 5 → 34%). In the shallowest water the straight line of eq. (5.1) is also least exact, because of internal reflection (Section 3). Its error estimate is too small (Section 5.2).

So `ATTENUATION` is kept, off unless asked, with no effect on Lyzenga's fit. K at 443–561 nm is the usable product. The attenuation depth is a diagnostic.

7 Lyzenga's calibration condition

7.1 The condition

Substituting the physical model (3.4) into the regression (2.1) gives (Lyzenga et al. 2006, eq. 11, in our signs)

$$\hat{Z} = \Big(a_0 + \sum_m a_m (B + c_m)\Big) - \Big(\sum_m a_m g_m\Big) Z. \tag{7.1}$$

The estimate is unbiased when the offset vanishes and the *gain* $G = -\sum_m a_m g_m$ is 1 (their eq. 12). With N bands there are $N + 1$ coefficients, so the fit has $N - 1$ degrees of freedom to spare; it can stay calibrated across several bottoms or waters. A change of a spectrally flat bottom brightness B leaves the depth unchanged if $\sum_m a_m = 0$ (their eq. 13, with the same specific attenuation in every band).

Lyzenga et al. do not impose eq. 12. They fit by unconstrained least squares (their eqs. 14 and 15), as *photic* does, and least squares meets it only on average over the soundings.

With $\ln L$. `LSM 0` uses $X_m = \ln L_m$, not $\ln(L_m - L_{\infty,m})$. Its derivative with depth is $-g_m(L_m - L_{\infty,m})/L_m$, which falls to zero as the signal approaches deep water. The local gain is then

$$G_{\text{eff}}(\mathbf{x}, Z) = -\sum_m a_m \, g_m(\mathbf{x}) \, \frac{L_m - L_{\infty,m}}{L_m}, \tag{7.2}$$

and it falls with depth whatever the coefficients.

7.2 What the chosen fit does

For the joint model ($\ln L$, four bands, `POWER`), the stage-1 attenuation gives the gain directly (Table 3).

- **The fit is calibrated only where the soundings are dense (5–10 m).**
- **The saturation near 16 m is eq. 12 failing:** the predicted gain falls to 0.27 at 15–20 m, against an observed slope of 0.18. The K maps predict it with no reference to the test depths.

Table 3: The gain of eq. (7.1) predicted from the stage-1 attenuation, and the observed slope of the fitted depth against ICESat-2 (test half, by depth bin).

	0–2 m	2–5 m	5–10 m	10–15 m	15–20 m
Predicted gain G_{eff}	1.32	1.08	**1.00**	0.77	0.27
Observed slope	0.05	0.65	**1.00**	0.90	0.18

- **Two departures are explained:** `POWER` stretches the depths beyond 6.9 m, which is why 10–15 m is observed above the prediction. Over a 2 m bin, the reference's own scatter dilutes the observed slope, so the shallow bins are not comparable.
- **The gain also varies across the box,** from 1.30 to 1.84 (10th to 90th percentile) with the nominal attenuation, because K varies 1.5–2× and one set of coefficients cannot follow it. At 10 m, a gain off by 15% is a 1.5 m error.
- **The fit is nearly invariant to a flat brightness change:** $\sum_m a_m = -1.27$, so a 10% brighter bottom moves the depth by 0.12 m. It works through large opposing blue and green coefficients (band sums +25.6 and −21.7), as Lyzenga et al.'s (their $h_1 \approx -h_2$), at the cost of noise amplification ($\sum |a_m| = 102$).

7.3 Imposing it

Two options use the stage-1 attenuation, which now runs before the fit.

- `CONSTRAIN GAIN` imposes $-\sum_m a_m \bar{g}_m = 1$, where $\bar{g}_m$ is the mean attenuation over the fit's soundings. `BRIGHTNESS` adds $\sum_m a_m = 0$.
- `NORMALISE` divides each predictor by the local $g_m(\mathbf{x})$. Then $X_m/g_m = (B + c_m)/g_m - Z$, so the gain is $-\sum_m a_m$ at *every* cell, whatever the attenuation there. With `CONSTRAIN GAIN` the condition becomes $\sum_m a_m = -1$, and `BRIGHTNESS` adds $\sum_m a_m/\bar{g}_m = 0$.

The constraints are linear equalities $C\mathbf{a} = \mathbf{d}$, none involving a_0. They are solved exactly: the constraint matrix is reduced to row echelon form, one coefficient per constraint is eliminated, and the reduced problem is fitted with the same Householder QR and Huber weights. With $\mathbf{a} = T\mathbf{a}_r + \mathbf{t}$, the covariance for `ERROR` is $C = TC_rT^{\mathsf{T}}$. `make test` checks that both constraints hold to four decimals.

Both options want $\ln(L - L_\infty)$: with $\ln L$, the effective attenuation changes with depth, so a fixed constraint is mis-specified.

7.4 Results

Out of fold on the Gulf's calibration half (the same four folds that chose the model; Table 4). Subtracting L_∞ turns the optically-deep test on, which removes 4–7% of the cells. Nearly all are deeper than 10 m, and with four bands it is mostly the red's doing. A missing cell counts as a miss, so the table also scores every variant on the cells where all of them have a depth.

- **Imposing eq. 12 changes almost nothing** (−1.4 to +0.4 points). Once L_∞ is subtracted, the free fit already meets it on average (with `NORMALISE`, the free fit's gain at the soundings is 0.92–1.01), and the constraint does as well as `POWER`. Regression dilution is not the main error.

Table 4: Lyzenga's calibration condition imposed on the joint fit, four bands, out of fold (5,833 cells; "same cells": the 5,201 where every variant has a depth).

	Hits	With a depth	Same cells	RMSE (m)	10–20 m
$\ln L$, `POWER` (chosen)	**83.9%**	96.3%	89.2%	1.60	92%
$\ln L$, `CONSTRAIN GAIN`	83.1%	96.3%	88.5%	1.59	90%
$\ln(L - L_\infty)$, free	77.5%	89.2%	86.9%	1.95	79%
$\ln(L - L_\infty)$, `POWER`	77.6%	89.2%	87.0%	1.99	80%
$\ln(L - L_\infty)$, `CONSTRAIN GAIN`	77.9%	89.2%	87.4%	1.90	83%
with `BRIGHTNESS`	76.2%	89.2%	85.4%	1.94	81%
`NORMALISE`	79.9%	89.2%	**89.6%**	1.78	91%
`NORMALISE CONSTRAIN GAIN`	79.5%	89.2%	89.2%	1.79	91%

- `BRIGHTNESS` **hurts every time,** by 1.5–5 points. The bottom is not spectrally flat, and the free fit uses how bottom colour changes with depth at this site.
- `NORMALISE` **is the real effect.** On the same cells it gains 2.7 points over the same $\ln(L - L_\infty)$ fit with four bands, and 3.0 with 443/483/561 nm, nearly all at 10–20 m (79 → 91% and 80 → 93%). So the spatial variation of K is the dominant error at depth. With 443/483/561 nm it is the best fit overall: 82.2% against $\ln L$'s 78.4%.
- **With four bands it only ties the chosen model** on the same cells (89.6 against 89.2%), and loses overall on coverage. Loosening the deep test (`DEEP 1`, `DEEP 0`) recovers coverage but costs more accuracy than it gains.

So the chosen model is unchanged, and `CONSTRAIN` and `NORMALISE` are kept as options, off by default. `NORMALISE` is how per-site K maps enter an empirical depth. Two next steps are possible: normalise $\ln L$ by its effective attenuation $g_m(L_m - L_{\infty,m})/L_m$, which keeps $\ln L$'s coverage; or fill the optically deep cells from the $\ln L$ fit.

8 Depth-invariant bottom indices

Lyzenga (1978) rotates the N log-bands of one scene into $N - 1$ indices that do not depend on depth, and one that does (his eqs. 8 and 9). Within a scene, $X_j = a_j - b_j Z$, with a_j depending on the bottom and $b_j = g_j$. The rotation takes Y_N along $\mathbf{b}$, so $Y_N = B_m - CZ$ carries the depth, and the other axes orthogonal to it, so they carry only the bottom. His Appendix B gives the rotation:

$$Y_i = \sum_{j=1}^{N} A_{ij} X_j, \qquad A_{ij} = \frac{b_{i+1}\, b_j}{\sqrt{S_i S_{i+1}}} \; (j \le i), \quad A_{i,i+1} = -\sqrt{\frac{S_i}{S_{i+1}}}, \quad A_{ij} = 0 \; (j > i+1), \tag{8.1}$$

for $i = 1, \ldots, N-1$, where $S_i = \sum_{k \le i} b_k^2$. With two bands this is the familiar index $(b_2 X_1 - b_1 X_2)/|\mathbf{b}|$, which `COMPUTE BOTTOM` already provides with a fixed attenuation ratio.

`INDICES` (with `ATTENUATION`) computes them for every scene, with $b_j = g_j(\mathbf{x})$ from stage 1. So the rotation follows the water across the scene, rather than assuming one attenuation ratio everywhere. The outputs are `/prefix/_index/i/_s/scene/`. `make test` checks them against the exact values on synthetic data, where the depth term cancels exactly (largest error 2.3×10^{-4}).

At the Gulf (Figure 4). Each scene's index carries that scene's constants ($\sum_j A_{ij}c_j$: the bands' bottom colour, deep-water and atmospheric terms), so the maps show anomalies from each scene's median. Two checks follow.

- **Most of the depth is removed, but not all.** A band's own $\ln(L - L_\infty)$ has $r = -0.80$ to -0.64 with the ICESat-2 depth. With 443/483/561 nm, index 1 has $r = -0.28$ to $+0.14$, and index 2 still $+0.31$ to $+0.51$. Index 2 shows the same near-shore against offshore split in every scene. There are three candidates: errors in K; a bottom that really changes with depth at this site; and the apparent reflectance of Section 3, which depends on the water.
- **The scenes do not agree.** The bottom is the same in every scene, so its indices should be too, but the median correlation between scenes is only 0.17 (index 1) and 0.38 (index 2); 0.19–0.63 with four bands. Index 1 is dominated by large patterns that differ between scenes: the atmosphere, adjacency and the water.

So the rotation is right, but at this site the bottom's contrast is small next to these effects, and the indices are not yet a bottom map. They stay, off by default, as the formalism for two bottom classes in stage 2 ($Y_N = B_m - CZ$ with a class-specific B_m, which is Lyzenga's own route to the depth) and as features for the learned model.

9 Usage

```
MODEL lyzenga_log_linear GRIDS g1 ... gn BATHYMETRY H [options, any order]
  LSM l1 ... ln | PERCENTILE p    DEEP k    RADIUS r    DEPTHS zmin zmax
  TIDE h    HUBER k    BALANCE bin    POWER [ALL]    ERROR H_error [NOISE] [GAIN]
  ATTENUATION K SCENES S [TIDES h1 ... hS] [SUNZENITH t1 ... tS]
                         [VIEWZENITH t1 ... tS] [NODES deg] [SMOOTHING w]
  CONSTRAIN GAIN [BRIGHTNESS]    NORMALISE    INDICES
                              (these, and ERROR ... GAIN, need ATTENUATION)
```

`SCENES` must come before the per-scene lists, and the grids are scene-major. The outputs are:

- `H` and `H_error`;
- with `ATTENUATION`, `K_/grid/` for every grid, and `K_depth`, `K_depth_sigma` and `K_bottom`; with `INDICES`, `K_index/i/_s/scene/`;
- the files `model.bam` (the fitted model as applied), `output.csv` (every sounding, its model depth and whether it was used) and `attenuation.csv`.

At the Gulf, `validation/gulf_lyzenga.py` runs it. That script prepares the inputs, chooses the settings by cross-validation, scores the results and draws the figures; the modelling is all in *photic*. `make test` checks the fit and the attenuation on synthetic data that follow the models exactly:

- `tests/lyz_*`: the coefficients and every sounding's depth;
- `tests/att_*`: K at every node, to 3×10^{-6}, the depth at every cell, within 5 mm (the float precision of the band grids), and the depth-invariant indices, within 2×10^{-3}; then both constraints of `NORMALISE CONSTRAIN GAIN BRIGHTNESS`, exactly.

References

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- Pope, R. M. and Fry, E. S. (1997). Absorption spectrum (380–700 nm) of pure water. II. Integrating cavity measurements. *Applied Optics* 36(33), 8710–8723.

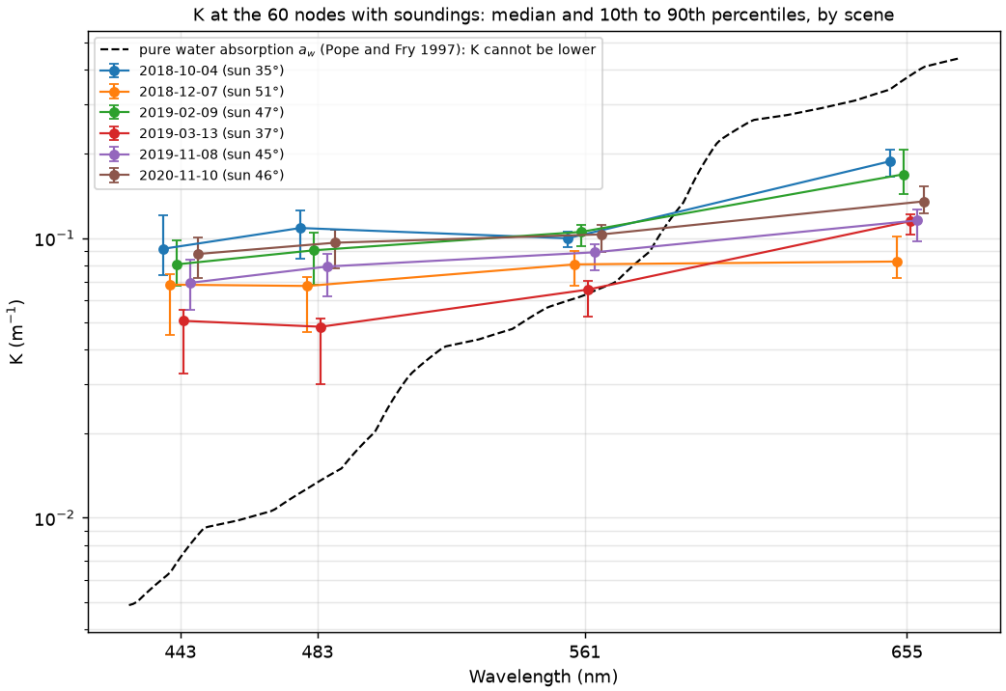


Figure 2: K of each scene against wavelength at the 60 nodes with soundings (median, with the 10th to 90th percentiles), and pure water's absorption a_w. K cannot physically fall below a_w.

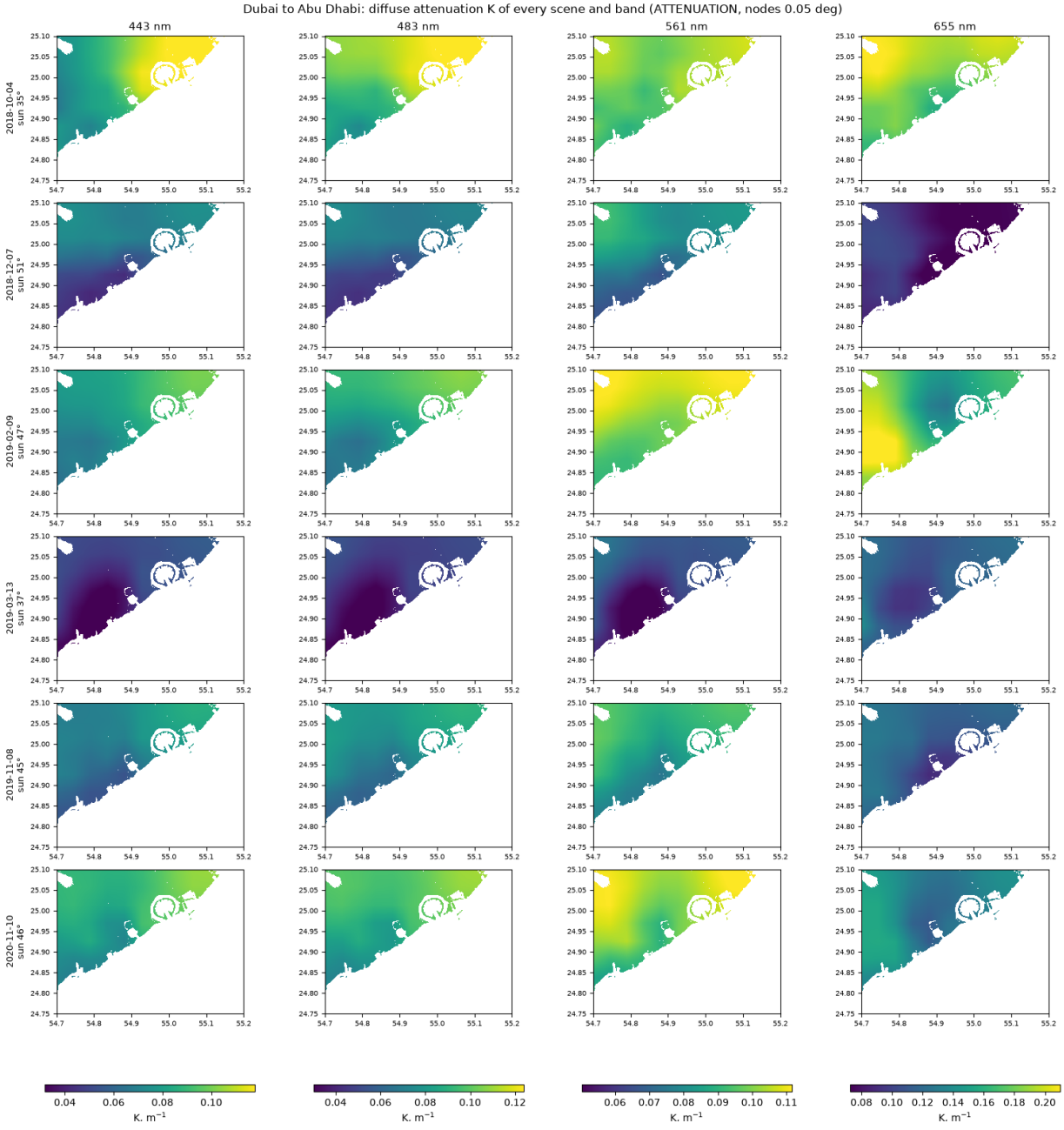


Figure 3: K of every scene (rows, with the date and sun zenith) and band (columns), on one colour scale per band. Nodes are 0.05° apart.

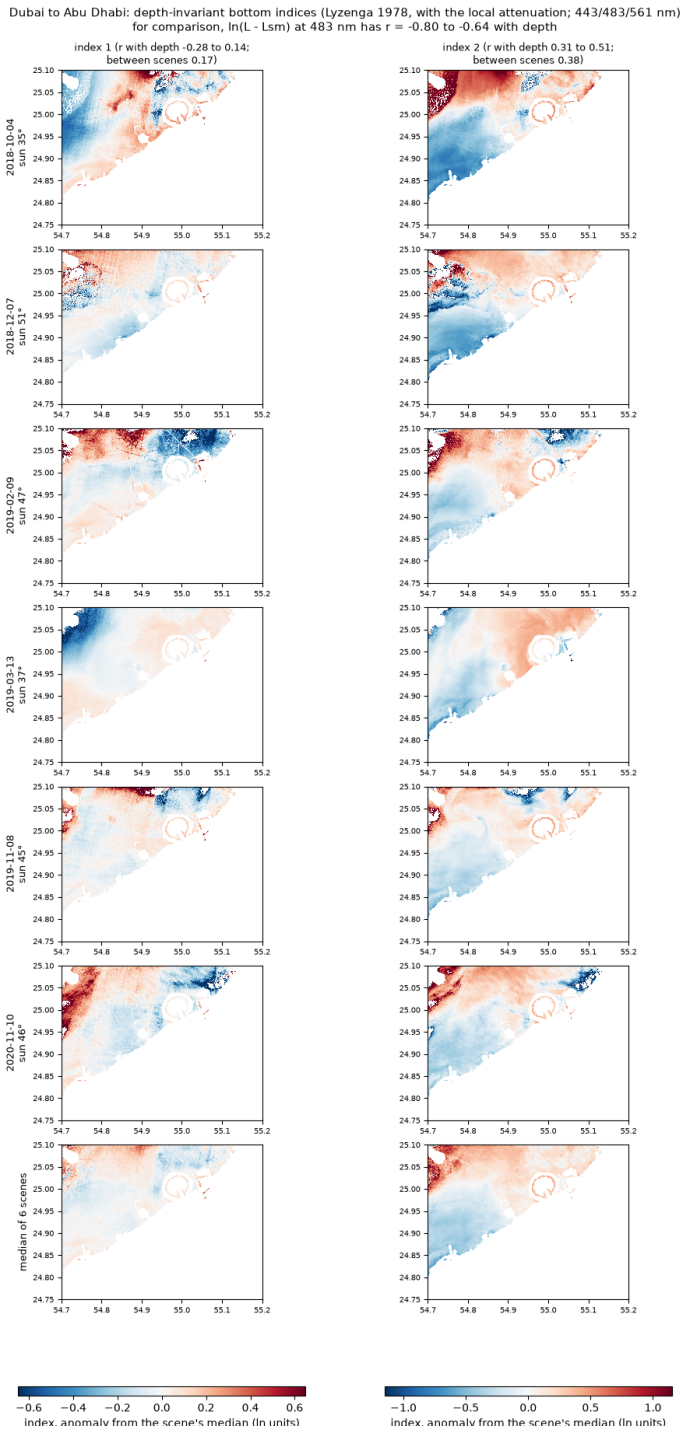


Figure 4: Depth-invariant bottom indices of every scene (443/483/561 nm, with the local attenuation), as anomalies from each scene's median, and their median over the scenes. The titles give each index's correlation with the ICESat-2 depth (six scenes) and between scenes (median).